

Mechanics

Lecture 3

Lagrange's Equations

(Goldstein Chapter 1)

Hamilton's Principle

(Chapter 2)

What We Did Last Time

- Discussed multi-particle systems
 - Internal and external forces
 - Laws of action and reaction
 - Introduced constraints
 - Generalized coordinates
 - Introduced Lagrange's Equations
 - ... and didn't do the derivation
- Let's pick it up and start from there

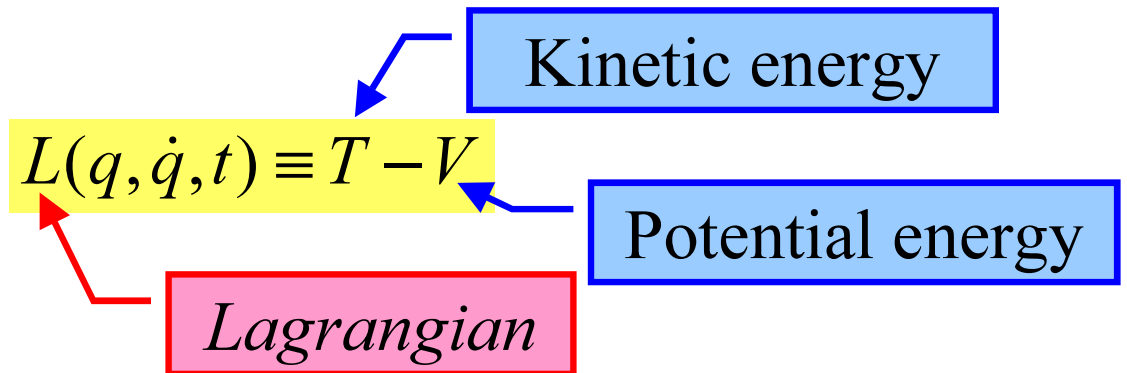
Today's Goals

- Derive Lagrange's Eqn from Newton's Eqn
 - Use D'Alembert's principle
 - There will be a few assumptions
 - Will make them clear as we go
- Introduce Hamilton's Principle
 - Equivalent to Lagrange's Equations
 - Which in turn is equivalent to Newton's Equations
 - Does not depend on coordinates by construction
 - Derivation in the next lecture

Lagrange's Equations

Recipe

$$\frac{d}{dt} \left(\frac{\partial L}{\partial \dot{q}_j} \right) - \frac{\partial L}{\partial q_j} = 0$$



- Express $L = T - V$ in terms of generalized coordinates $\{q_j\}$, their time-derivatives $\{\dot{q}_j\}$, and time t
 - The potential $V = V(q, t)$ must exist
 - i.e. all forces must be conservative

Virtual Displacement

- Consider a system with constraints

- Ordinary coordinates $\mathbf{r}_i$ ($i = 1 \dots N$)
- Generalized coordinates q_j ($j = 1 \dots n$)

$$\begin{cases} \mathbf{r}_1 = \mathbf{r}_1(q_1, q_2, \dots, q_n, t) \\ \mathbf{r}_2 = \mathbf{r}_2(q_1, q_2, \dots, q_n, t) \\ \vdots \\ \mathbf{r}_N = \mathbf{r}_N(q_1, q_2, \dots, q_n, t) \end{cases}$$

- Imagine moving all the particles

slightly $\mathbf{r}_i \rightarrow \mathbf{r}_i + \delta\mathbf{r}_i$ $q_j \rightarrow q_j + \delta q_j$

Virtual displacement

- Note that $\delta\mathbf{r}_i$ must satisfy the constraints

$$\delta\mathbf{r}_i = \sum_j \frac{\partial \mathbf{r}_i}{\partial q_j} \delta q_j$$

$3N$ coordinates
not independent

n coordinates
independent

D'Alembert's Principle

- From Newton's Equation of Motion

$$\mathbf{F}_i = \dot{\mathbf{p}}_i \longrightarrow \mathbf{F}_i - \dot{\mathbf{p}}_i = 0$$

- Part of the force $\mathbf{F}_i$ must be due to constraints

$$\mathbf{F}_i = \mathbf{F}_i^{(a)} + \mathbf{f}_i$$

“applied” force

“constraint” force

- Applied force is “known” $\mathbf{F}_i^{(a)} = \mathbf{F}_i^{(a)}(\mathbf{r}_1, \mathbf{r}_2, \dots, \mathbf{r}_i, \dots, \mathbf{r}_N, t)$
- Constraint force $\mathbf{f}_i$ (usually) does no work
 - Movement is perpendicular to the force $\mathbf{f}_i \delta \mathbf{r}_i = 0$
 - Exception: friction
- Now multiply $\mathbf{F}_i^{(a)} + \mathbf{f}_i - \dot{\mathbf{p}}_i = 0$ by $\delta \mathbf{r}_i$ and sum over i

D'Alembert's Principle

$$\sum_i (\mathbf{F}_i^{(a)} - \dot{\mathbf{p}}_i) \delta \mathbf{r}_i = 0$$

“constraint” force is out of the game.
You can forget (a)

- Force of constraints dropped out because $\mathbf{f}_i \delta \mathbf{r}_i = 0$
- Called D'Alembert's Principle (1743)
- Now we switch from $\mathbf{r}_i$ to q_j

$$\text{1st term} = \sum_i \mathbf{F}_i \sum_j \frac{\partial \mathbf{r}_i}{\partial q_j} \delta q_j = \sum_j Q_j \delta q_j$$

$$Q_j \equiv \sum_i \mathbf{F}_i \frac{\partial \mathbf{r}_i}{\partial q_j}$$

Generalized force

- Unit of Q_j not always [force]
- $Q_j q_j$ is always [work]

D'Alembert's Principle

$$\text{2nd term} = \sum_i \dot{\mathbf{p}}_i \delta \mathbf{r}_i = \sum_i \dot{\mathbf{p}}_i \sum_j \frac{\partial \mathbf{r}_i}{\partial q_j} \delta q_j = \sum_{i,j} m_i \ddot{\mathbf{r}}_i \frac{\partial \mathbf{r}_i}{\partial q_j} \delta q_j$$

■ A bit of work can show $\ddot{\mathbf{r}}_i \frac{\partial \mathbf{r}_i}{\partial q_j} \rightarrow \frac{d}{dt} \left[\frac{\partial}{\partial \dot{q}_j} \left(\frac{v_i^2}{2} \right) \right] - \frac{\partial}{\partial q_j} \left(\frac{v_i^2}{2} \right)$

$$= \sum_j \left\{ \frac{d}{dt} \left(\frac{\partial T}{\partial \dot{q}_j} \right) - \frac{\partial T}{\partial q_j} \right\} \delta q_j$$

$$T \equiv \sum_i \frac{mv_i^2}{2}$$

■ D'Alembert's Principle becomes

$$\sum_j \left\{ \left[\frac{d}{dt} \left(\frac{\partial T}{\partial \dot{q}_j} \right) - \frac{\partial T}{\partial q_j} \right] - Q_j \right\} \delta q_j = 0$$

Lagrange's Equations

$$\sum_j \left\{ \left[\frac{d}{dt} \left(\frac{\partial T}{\partial \dot{q}_j} \right) - \frac{\partial T}{\partial q_j} \right] - Q_j \right\} \delta q_j = 0$$

These are free

- Generalized coordinates q_j are independent

$$\frac{d}{dt} \left(\frac{\partial T}{\partial \dot{q}_j} \right) - \frac{\partial T}{\partial q_j} = Q_j$$

Almost there!

- Assume forces are conservative $\mathbf{F}_i = -\nabla_i V$

$$Q_j \equiv \sum_i \mathbf{F}_i \frac{\partial \mathbf{r}_i}{\partial q_j} = - \sum_i \nabla_i V \frac{\partial \mathbf{r}_i}{\partial q_j} = - \frac{\partial V}{\partial q_j}$$

Throw this
back in

Lagrange's Equations

$$\frac{d}{dt} \left(\frac{\partial T}{\partial \dot{q}_j} \right) - \frac{\partial (T - V)}{\partial q_j} = 0$$

■ Assume that V does not depend on $\dot{q}_j$ $\rightarrow \frac{\partial V}{\partial \dot{q}_j} = 0$

Finally 

$$\frac{d}{dt} \left(\frac{\partial L}{\partial \dot{q}_j} \right) - \frac{\partial L}{\partial q_j} = 0$$

$$L = T(q_j, \dot{q}_j, t) - V(q_j, t)$$

Done!

Assumptions We Made

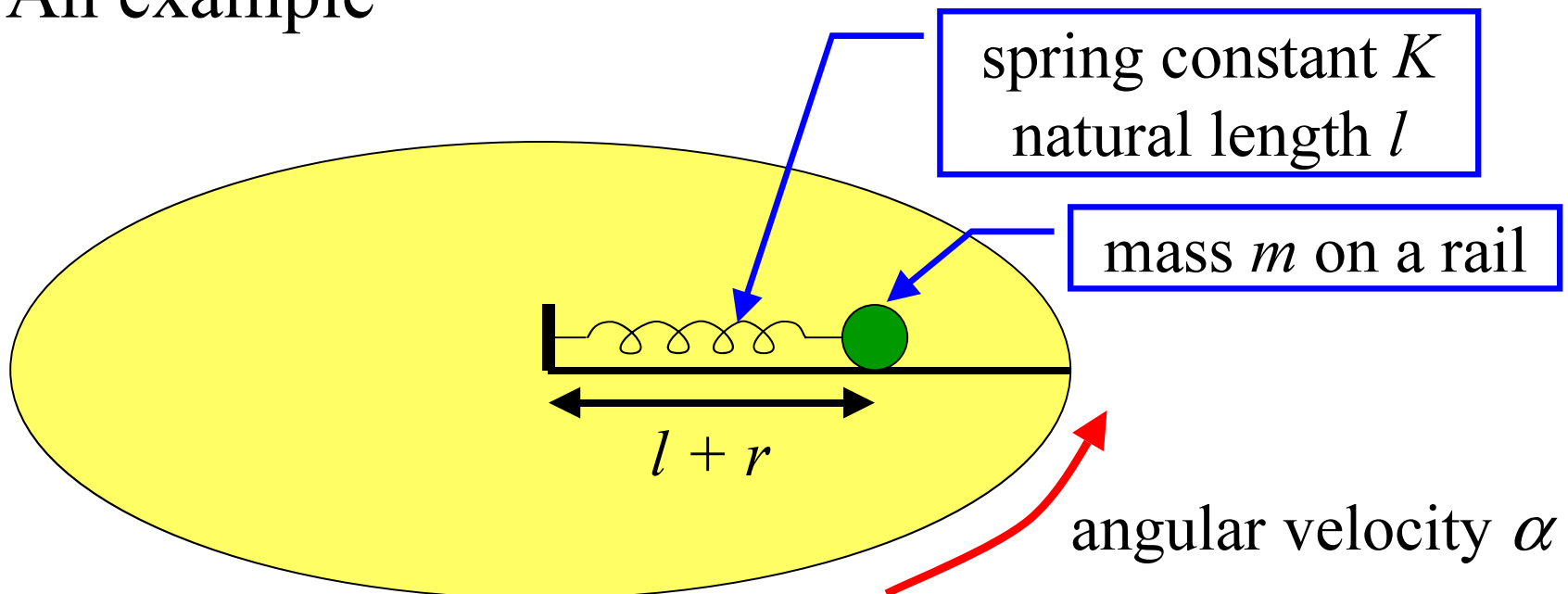
- Constraints are holonomic $\Rightarrow \mathbf{r}_i = \mathbf{r}_i(q_1, q_2, \dots, q_n, t)$
 - We always assume this
- Constraint forces do no work $\Rightarrow \mathbf{f}_i \delta \mathbf{r}_i = 0$
 - Forget frictions
- Applied forces are conservative $\Rightarrow \mathbf{F}_i = -\nabla_i V$
 - Lagrange's Eqn. itself is OK if V depends explicitly on t
- Potential V does not depend on $\dot{q}_j$ $\Rightarrow \frac{\partial V}{\partial \dot{q}_j} = 0$

Will review the last assumption later

Example: Time-Dependent

- Transformation functions may depend on t
 - Generalized coordinate system may move
 - E.g. coordinate system fixed to the Earth
- An example

$$\mathbf{r}_i = \mathbf{r}_i(q_j, t)$$



Example: Time-Dependent

- Transformation functions:

$$\begin{cases} x = (l + r) \cos \alpha t \\ y = (l + r) \sin \alpha t \end{cases}$$

- Kinetic energy

$$T = \frac{m}{2} \{ \dot{x}^2 + \dot{y}^2 \} = \frac{m}{2} \{ \dot{r}^2 + (l + r)^2 \alpha^2 \}$$

- Potential energy

$$V = \frac{K}{2} r^2$$



$$L = \frac{m}{2} \{ \dot{r}^2 + (l + r)^2 \alpha^2 \} - \frac{K}{2} r^2$$

Lagrange's Equation

$$\frac{d}{dt} \left[\frac{\partial L}{\partial \dot{r}} \right] - \frac{\partial L}{\partial r} = m\ddot{r} - m\alpha^2(l + r) + Kr = 0$$

Example: Time-Dependent

$$\frac{d}{dt} \left[\frac{\partial L}{\partial \dot{r}} \right] - \frac{\partial L}{\partial r} = m\ddot{r} - m\alpha^2(l+r) + Kr = 0$$

$$m\ddot{r} + (K - m\alpha^2) \left(r - \frac{m\alpha^2 l}{K - m\alpha^2} \right) = 0$$

- If $K > m\alpha^2$, a harmonic oscillator with $\omega = \sqrt{\frac{K - m\alpha^2}{m}}$
 - Center of oscillation is shifted by
- If $K < m\alpha^2$, moves away exponentially
- If $K = m\alpha^2$, velocity is constant
 - Centripetal force balances with the spring force

Note on Arbitrariness

- Lagrangian is **not unique** for a given system

- If a Lagrangian L describes a system

$$L' = L + \frac{dF(q,t)}{dt} \quad \text{works as well for any function } F$$

- One can prove

$$\frac{d}{dt} \left(\frac{\partial}{\partial \dot{q}} \left(\frac{dF}{dt} \right) \right) - \frac{\partial}{\partial q} \left(\frac{dF}{dt} \right) = 0 \quad \text{using} \quad \frac{dF}{dt} = \frac{\partial F}{\partial q} \dot{q} + \frac{\partial F}{\partial t}$$

Hamilton's Principle

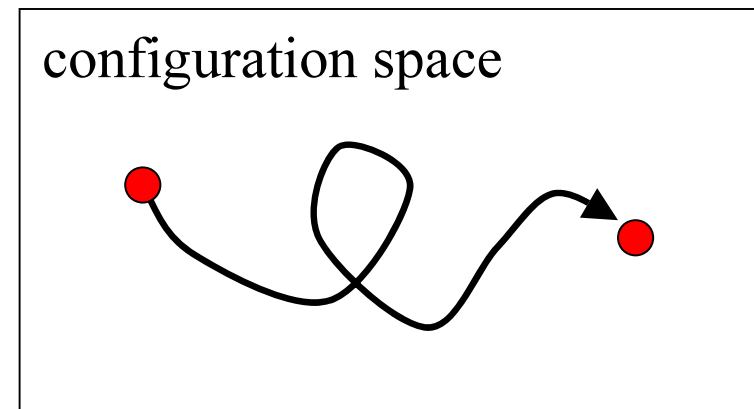
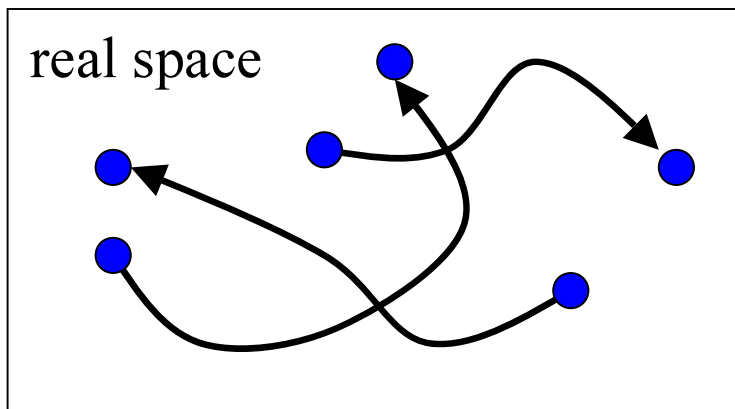
- We derived Lagrange's Eqn from Newton's Eqn using a “differential principle”
 - D'Alembert's principle uses infinitesimal displacements
- It's possible to do it with an “integral principle”

Hamilton's Principle

Configuration Space

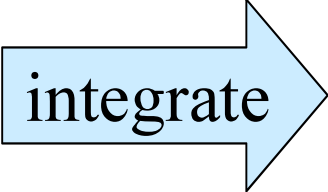
- Generalized coordinates $q_1, \dots, q_n$ fully describe the system's **configuration** at any moment
- Imagine an n -dimensional space
 - Each point in this space ($q_1, \dots, q_n$) corresponds to one configuration of the system
 - Time evolution of the system $\rightarrow$ A curve in the configuration space

configuration space



Action Integral

- A system is moving as $q_j = q_j(t) \quad j = 1 \dots n$
- Lagrangian is $L(q, \dot{q}, t) = L(q(t), \dot{q}(t), t)$

 $I = \int_{t_1}^{t_2} L dt$ Action, or action integral

- Action I depends on the entire path from t_1 to t_2
- Choice of coordinates q_j does not matter
 - Action is invariant under coordinate transformation

Hamilton's Principle

The action integral of a physical system is *stationary* for the actual path

- This is equivalent to Lagrange's Equations

- We will prove this

We will also define “stationary”

- Three equivalent formulations

- Newton's Eqn depends explicitly on *x-y-z coordinates*

- Lagrange's Eqn is same for any *generalized coordinates*

- Hamilton's Principle refers to *no coordinates*

- Everything is in the action integral

Hamilton's Principle is more fundamental

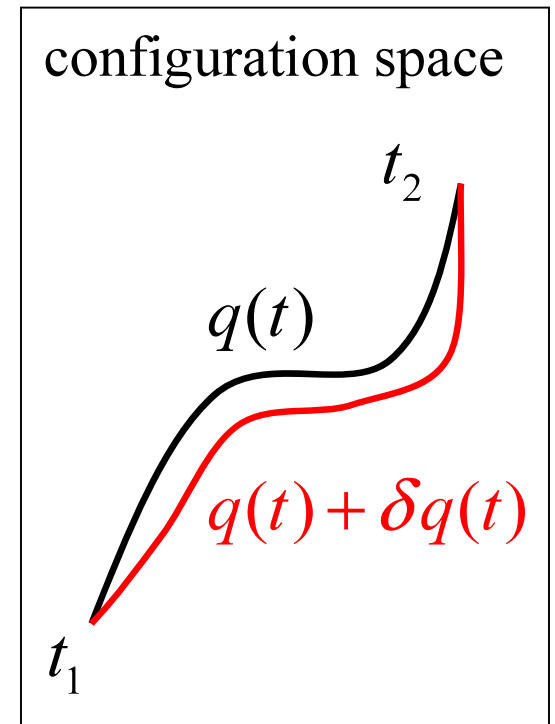
probably...

Stationary

- Consider two paths that are close to each other
 - Difference is infinitesimal
- **Stationary** means that the difference of the action integrals is zero to the 1st order of $\delta q(t)$
 - Similar to “first derivative = 0”

$$\delta I = \int_{t_1}^{t_2} L(q + \delta q, \dot{q} + \delta \dot{q}, t) dt - \int_{t_1}^{t_2} L(q, \dot{q}, t) dt = 0$$

- Almost same as saying “minimum”
 - It could as well be maximum



$$\delta q(t_1) = \delta q(t_2) = 0$$

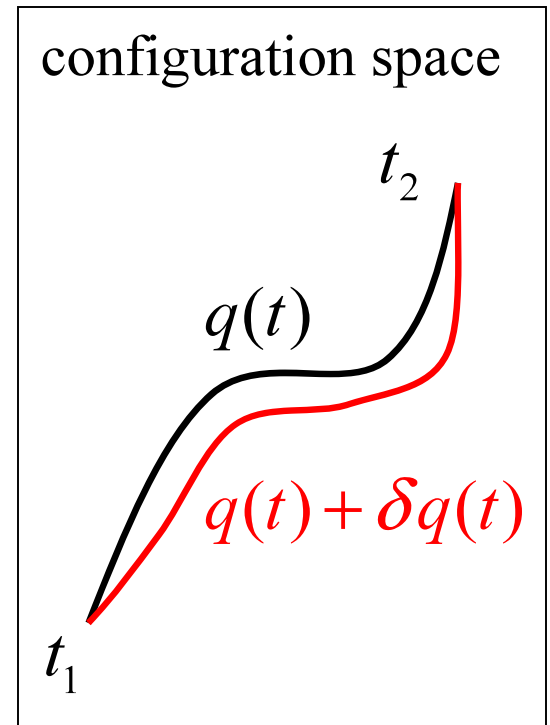
Infinitesimal Path Difference

■ What's $\delta q(t)$?

- It's arbitrary ... sort of
- It has to be zero at t_1 and t_2
- It's **well-behaving**

Continuous, non-singular,
continuous 1st and 2nd derivatives

Don't worry
too much



■ Have to shrink it to zero

- Trick: write it as $\delta q(t) = \alpha \eta(t)$
 - α is a parameter, which we'll make $\rightarrow 0$
 - $\eta(t)$ is an arbitrary **well-behaving** function $\eta(t_1) = \eta(t_2) = 0$

Hamilton $\rightarrow$ Lagrange

- To derive Lagrange's Eqns from Hamilton's Principle

$$\delta I = \int_{t_1}^{t_2} L(q + \delta q, \dot{q} + \delta \dot{q}, t) dt - \int_{t_1}^{t_2} L(q, \dot{q}, t) dt = 0$$

- Define $I(\alpha) \equiv \int_{t_1}^{t_2} L(q(t) + \alpha \eta(t), \dot{q}(t) + \alpha \dot{\eta}(t), t) dt$

- δI is then $\lim_{\alpha \rightarrow 0} [I(\alpha) - I(0)] \rightarrow \left(\frac{\partial I}{\partial \alpha} \right)_{\alpha=0} d\alpha$

- We must show that $\left(\frac{\partial I}{\partial \alpha} \right)_{\alpha=0} = 0$ leads to Lagrange's Eqns

A bit of work. Will do it on Thursday

Summary

- Derived Lagrange's Eqn from Newton's Eqn
 - Using D'Alembert's Principle ← Differential approach
- Assumptions we made:
 - Constraints are holonomic → Generalized coordinates
 - Forces of constraints do no work → No frictions
 - Other forces are monogenic → Generalized potential
- Introduced Hamilton's Principle
 - Integral approach
 - Defined the action integral and “stationary”
 - Derivation in the next lecture

$$Q_j = -\frac{\partial U}{\partial q_j} + \frac{d}{dt} \left(\frac{\partial U}{\partial \dot{q}_j} \right)$$